SILVIA ROSSI

Senior Scientist

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I am a senior scientist (former ERCIM Fellow) in the Distributed and Interactive Systems (DIS) group at Centrum Wiskunde & Informatica (CWI). My research focuses on user behaviour metrics and analysis for immersive and interactive media, from single viewers to social experiences. My expertise lies at the intersection of multimedia systems, predictive machine learning and human-centred computing, with a proven track record of bridging behavioural science with system design and quality-of-experience evaluation.

SELECTED PUBLICATIONS

- **S. Rossi**, C. Ozcinar, A. Smolic, and L. Toni "*Do Users Behave Similarly in VR? Investigation of the User Influence on the System Design.*" ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM), 2020. doi:10.1145/3381846.
  [*2021 Nicholas Georganas ACM TOMM Best Paper Award*].
- X. Zhou, I. Viola, **S. Rossi**, and P. Cesar. "*Comparison of Visual Saliency for Dynamic Point Clouds: Task-free vs. Task-dependent*" in IEEE Transactions on Visualization and Computer Graphics, 2025.
- S. Lee, I. Viola, **S. Rossi**, Z. Guo, I. Reimat, K. Ławicka, A. Striner, P. Cesar. "*Designing and Evaluating a VR Lobby for a Socially Enriching Remote Opera Watching Experience.*" in IEEE Transactions on Visualization and Computer Graphics, 2024. doi:10.1109/TVCG.2024.3372081.

WORK EXPERIENCE

Senior Scientist — Amsterdam, The Netherlands
CWI - Distributed and Interactive System (DIS) group — *May 2024 - Present*

- Established and lead the research on behavioural analysis, developing an interdisciplinary framework that integrates social science methodologies with multimedia systems; directing the development of new metrics and methodologies for behavioural analysis in immersive systems; leading task-level objectives within EU Horizon projects (TRANSMIXR, INDUX-R), and managing cross-consortium tasks; supervising technical staff and PhD/Master students, fostering a collaborative environment.

Postdoctoral Researcher (including ERCIM Fellow) — Amsterdam, The Netherlands
CWI - Distributed and Interactive System (DIS) group — *May 2022 - April 2024*

- Developed predictive models and methodologies to analyze user behavioral similarities in 6-DoF immersive media (e.g., point clouds), optimizing real-time multimedia performance; worked on immersive communication frameworks to enable high-fidelity Social VR experiences, focusing on the intersection of user experience (UX) and system optimisation.

Research Internship — Amsterdam, The Netherlands
CWI - Distributed and Interactive System (DIS) group — *October 2020 – March 2021*

- Internship on user behavioural analysis in immersive system from 3-DoF to 6-DoF.

Research Internship — London, United Kingdom
UCL - Department of Electronic and Electrical Engineering — *November 2016 - April 2017*

- Research on interactive communication: optimisation of VR content streaming system.

Visiting Research Kingston upon Thames, United Kingdom
Kingston University - WMN research group *September 2016 – October 2016*

- Further research on master thesis ("Content characterisation of 3D videos").

Substitute Teacher Lugo (RA), Italy
ICT (Information and Communications Technology) Institute *May 2016 - June 2016*

- Teacher of 3 courses (physic, electronic and electrical) in year 3,4 and 5.

Visiting Research Kingston upon Thames, United Kingdom
Kingston University - WMN research group *September 2015 – February 2016*

- Period of exchanging research to conduct my master thesis project ("Content characterisation of 3D videos").

EDUCATION

University College London (UCL) London, United Kingdom
PhD in Electronic and Electrical Engineering *May 2017 - Feb 2022*

- *PhD project title*: Understanding user interactivity for the next-generation immersive communication: design, optimisation, and behavioural analysis
- *Supervisor*: Dr. Laura Toni.
- *Expertise and skills include*: mathematical modelling, computer-based simulations, interactive multimedia communication system, video processing, virtual reality, machine learning, optimisation problem.
- *Principal subjects of study*: development of human-centric and content-dependent streaming strategies for VR content; analysis of user behaviour during the navigation within immersive content to learn their movements through machine learning techniques and data analysis.

University of Bologna (Cesena Campus) Cesena, Italy
MSc in Electronic and Telecommunication Engineering for Energy *September 2013 - March 2016*

- *Grade*: 110/110.
- *Project title*: Content Characterisation of 3D Videos.
- *Supervisor*: Prof. Marco Chiani (University of Bologna (Cesena campus), Italy) and Prof. Maria Martini (Kingston University, UK).
- *Principal subjects of study*: 3D video content analysis, mathematical modelling of depth indicator, subjective test for video quality assessment.
- 5 month research period at Wireless Multimedia Networking (WMN) laboratory in Kingston University (Kingston upon Thames - UK) partially founded with a travel grant by University of Bologna.

University of Bologna (Cesena Campus) Cesena, Italy
BSc in Electronics, Telecommunications, and Information Engineering *Sep 2009 – Mar 2013*

- *Grade*: 98/110
- *Project title*: Elaborazione di segnali video per monitoraggio in situazioni di emergenza (Video signal processing for monitoring in emergency situation).
- *Supervisor*: Prof. Marco Chiani.
- *Principal subjects of study*: multi-view video database, simulations to test a video-camera selection algorithm, multi-view system optimisation.

- Part of the project (i.e., multi-view video database) has been used in researches founded by European project, FP7 Content and cOntext aware delivery for iNteraCtive multimedia healthcaRe applications (CONCERTO).

SCHOLARSHIPS & AWARDS

Best Paper Award 2025
International Conference on Quality of Multimedia Experience (QoMEX'25)

- Recognized for the article "From Individual QoE to Shared Mental Models: A Novel Evaluation Paradigm for Collaborative XR", which introduces a pioneering evaluation framework for Social XR by integrating psychology metric (Shared Mental Models) with multimedia systems, shifting the paradigm from individual QoE to group-level dynamics.

Best SIGMM PhD Thesis Award 2023
ACM Special Interest Group on Multimedia (SIGMM)

- The SIGMM Award for Outstanding PhD Thesis in Multimedia Computing, Communications and Applications is given once per year to a researcher whose PhD thesis has the potential of very high impact in multimedia computing, communication and applications, or gives direct evidence of such impact.

Fabrizio Lombardi Prize 2022
University College London (UCL)

- The Lombardi prize is awarded annually by UCL's Department of Electronic and Electrical Engineering to the student who submitted the best PhD thesis over the previous year.

ERCIM Alain Bensoussan Fellowship 2022
ERCIM

- Prestigious one-year postdoctoral fellowship for the project: "Behavioural analysis in 6-DoF VR system."

ACM TOMM Nicholas Georganas Best Paper Award 2021
ACM Transaction on Multimedia Computing, Communications and Applications (TOMM)

- Awarded for the most significant work published in ACM TOMM in the calendar year for the article: "Do Users Behave Similarly in VR? Investigation of the User Influence on the System Design". The paper introduced a novel user-affinity metric and an server optimization that achieved up to 70% savings in encoding and storage costs while maximizing user experience. It also contributed a widely-used public dataset of navigation trajectories across heterogeneous devices, advancing the state-of-the-art in user-centric VR system design.

Mobility Grant 2021
Royal Society

- 6-month internship as part of the exchange project between UCL and CWI named "User-centric omnidirectional video streaming for VR"
  Royal Society under grant IES—R1—180128.

Graduate Research Scholarship 2017 - 2022
University College London (UCL)

- Fully funded scholarship for PhD degree.

Travel Grant to MMSys 2018 - 2019
ACM SIGMM

- June 2019, Amherst - MA, USA.

- June 2018, Amsterdam, Netherlands.

Sponsorship 2018

Institute of Communications and Connected Systems (ICCS) at UCL

- Attendance to the 57th FITCE Congress ("Delivery and Consumption of Digital Media") September 2018, MediaCityUK (Salford - UK).

Cisco Prize for the Best Student Poster, Barlow lecture 2018

University College London (UCL)

- Award for the best PhD student poster in the EEE department of UCL.

Travel Grant 2016

University of Bologna

- Partial funding for an exchange period abroad to conduct a master thesis project.

TECHNICAL SKILLS & PROFESSIONAL DEVELOPMENT

- *Programming and tools*: Python (pandas, NumPy, SciPy), deep learning (PyTorch, TensorFlow), MATLAB, ffmpeg, Git and GitHub, LaTeX, C/C++ (basic).
- *Professional development (Coursera, 2026)*: DeepLearning.AI PyTorch: Fundamentals; Google AI Fundamentals.

ACADEMIC SERVICES

EXECUTIVE COMMITTEE MEMBER

- **SIGMM Director of Conferences** November 2024-Present
  ACM Special Interest Group on Multimedia (SIGMM) - http://sigmm.org
  - Elected by the SIGMM community to the Executive Committee as Director of Conferences. The ACM Special Interest Group on Multimedia (SIGMM) is a forum for researchers, engineers, and practitioners in all aspects of multimedia computing, communication, storage, and applications. The main tasks are to supervise current SIGMM-sponsored conferences (e.g., ACM MM, ACM MMsys) in all aspects of the organisation, ensuring their success and smooth execution of the conference; monitor conference logistics and financial planning while serving as a member of the SIGMM Executive Board.
- **Steering Committee Chair of ACM Multimedia** November 2024 – Present
  ACM Multimedia (MM) https://acmmm.org/index.html
  - Lead the committee responsible for the long-term planning, policy decisions, and site selection for the ACM Multimedia conference series; manage the transition between yearly organizing teams and ensure the conference maintains its scientific standards and financial stability.
- **Memeber of SIGMM Strike Team on Harmonization and Spread** 2024-2026
  ACM Special Interest Group on Multimedia (SIGMM) - http://sigmm.org
  - The task of the team is to integrate SIGMM Records and Social Media in the whole process of the ACM Multimedia organisation, improve synchronisation and harmonisation between ACM Multimedia and other SIGMM Conferences, and strengthen the sense of community.
- **Information Director of SIGMM Records** 2021-Present
  ACM SIGMM Records - https://records.sigmm.org
  - SIGMM Records is the quarterly newsletter of ACM SIGMM for disseminating multimedia-related information, and its official announcement channel. As Information Director, I am responsible for ACM SIGMM social media channels (i.e., Twitter and LinkedIn) and helping the editor in chief with the publication of new multimedia related articles.

ORGANISING COMMITTEE MEMBER

- **General Co-Chair** of ACM International Conference on Interactive Media Experiences (IMX) 2027 (ACM IMX 2027)
- **Program Co-Chair** of ACM International Conference on Interactive Media Experiences (IMX) 2026 (ACM IMX 2026); ACM Multimedia Systems Conference (MMsys) 2025 (ACM MMSys 2025)
- **Workshop Chair** of IMmersive Mixed and Virtual Environment Systems (MMVE) 2024 (MMVE 2024)
- *Doctoral Symposium* Co-Chair of ACM Multimedia Systems Conference (MMsys) 2027 (ACM MMsys 2027)
- *Panel Co-Chair* of ACM Multimedia (MM) 2026 (ACM MM 2026)
- *Demo Co-Chair* of the conference on Quality of Multimedia Experience (QoMEX) 2025 (QoMEX 2025)
- *Work-in-Progress Co-Chair* of the conference on Interactive Media Experiences (IMX) 2025 (ACM IMX 2025)
- *Special Session Chair* of the conference on Quality of Multimedia Experience (QoMEX) 2024 (QoMEX 2024)
- *Social Media Chair* of ACM Multimedia Systems Conference (MMsys) from 2020 to 2022 (ACM MMSys 2020, ACM MMSys 2021, ACM MMSys 2022).

EDITORIAL

- Guest Editor, Special Issue on MMSys 2025 - ACM Transactions on Multimedia Computing, Communications and Applications (TOMM) (*in progress*)

AREA/ASSOCIATE CHAIR

- Area Chair of ACM Multimedia (2024-present)
- Associate Chair (ACM IMX) (2024-present)

REVIEWER

- **Journals**
  - ACM Transactions on Multimedia Computing, Communications and Applications (TOMM);
  - IEEE Transaction on Multimedia (website);
  - IEEE Journal of Selected Topics in Signal Processing (website).
- **Conferences (Technical Program Commitee member)**
  - ACM Multimedia (ACM MM) (since 2022);
  - ACM Multimedia Systems Conference (MMsys) (since 2023);
  - Quality of Multimedia Experience (QoMEX) (since 2023)
  - IEEE AIxVR (2024);
  - IEEE ICME (2020-2022).

- **Workshop**
  - IMmersive Mixed and Virtual Environment Systems (MMVE) (since 2021);
  - Methodologies for Multimedia (Meet4MM) (2024)
  - Interactive Extended Reality (IXR) (2023).

TEACHING & MENTORING

Co-organisers of the Spring School on Social XR 2023-Present
CWI, Amsterdam - https://www.dis.cwi.nl/spring-school/

- The aim of the school is to create a community of Social XR by welcoming students from technology, social sciences, and humanities and renowned experts on different research areas related in the field. In addition to talks by invited experts, there are networking events, including poster presentations specifically tailored to provide support and guidance for the participants. The school is intended for Ph.D. students who are interested in learning more about the mechanics underlying immersive experiences.
  Supported by SIGMM (Funding for Special Initiatives from 2023) and ACM Europe Seasonal School from 2025.

Student Supervision

- Co-supervisor of PhD student Fabrizio De Fiore (Technological University of the Shannon (TUS)) (2025 - Expected 2029) - Project title: "*HoloCalls: Understanding Behaviours in Communication-Centric Volumetric XR Telepresence System* "
- Daily supervisor (2024) of master student Varun Pradhan (TU Delft, Delft).
  Thesis: "*Predicting Head Pose in Virtual Reality*"
- Daily supervisor (2020-2022) of PhD student Pedro Gomes (UCL, London)
  Thesis: "*Graph-based Learning for Dynamic Point Cloud*"
- During my PhD at UCL (2017-2022), I have also supervised bachelor/master projects and thesis in the area of immersive media processing helping students in understanding and developing their research goals, teaching them software development practices and supporting them during their research path.

Teaching Assistant

- During my PhD at UCL, I have been a Postgraduate Teaching Assistants (PGTAs) for two courses related to signal processing and data analysis (e.g., Data Acquisition and Processing Systems and Data Analytics), and also a laboratory assistant for a consultancy course on machine learning. As part of my tasks in these experiences, I helped during laboratory sessions and also in building the laboratory exercises together with colleagues.

PROJECTS

- **INDUX-R (Grant Agreement no. 101135556)** 2024 - Present
  EU Horizon Europe - INDUX-R LinkedIn page

  - INDUX-R envisions a human-centric XR ecosystem that will transform European industrial sectors by empowering humans and creating innovate XR products and services of significant added value. My role in the project is to coordinate the activities of DIS group, leader of task 3.3. XR scene orchestration, resource scheduling and real-time synchronisation for multi-user XR applications, with the other partners in the project.
    CWI Budget: 460K Euros

- **TRANSMIXR (Grant Agreement no. 101070109)** 2023 - 2025
  EU Horizon Europe - https://transmixr.eu
  - TRANSMIXR will create a range of human-centric tools for remote content production and consumption via social virtual reality. My role as leader of Task Force on metrics is to identify and collect key state-of-art metrics and methodologies used for the evaluation of quality and user experience in XR.
    CWI Budget: 1M Euros

- **SPIRIT Open Call** 2025
  EU Horizon Europe - https://www.spirit-project.eu
  - The ambition of the SPIRIT project is to "research, develop, and demonstrate low-latency and scalable solutions that will ultimately bring real-time immersive telepresence into practice". In our open call project OPEN-DASH-PC, we propose the use of Ultra Low Latency Dynamic Adaptive Streaming over HTTP for Point Cloud contents (ULL-DASH-PC), an open-source transport protocol that enables multi-quality, multi-user real-time communication using point cloud volumetric representation. OPEN-DASH-PC aims to make the component available to the SPIRIT platform and to the scientific community, to foster further development in the field.
    CWI Budget: 150K Euros

GRANTS

- **ACM Europe Seasonal School** 2025-2026
  ACM Europe Council Seasonal Schools - https://europe.acm.org/seasonal-schools
  - Lead applicant for the successful bid to recognise the Spring School on Social XR at CWI as an officially ACM Europe Seasonal School. This prestigious designation identifies the school as a flagship European educational initiative, awarded based on demonstrated scientific excellence and its strategic contribution to the international research community.

- **SIGMM Funding for Special Initiatives** 2026-2023
  ACM Special Interest Group on Multimedia (SIGMM) - http://sigmm.org
  - Lead applicant for three consecutive successfully grants (totalling 39k Euros) to fund the Spring School on Social XR at CWI. These provided travel support for early-career researchers and expanded the school's international reach by hosting speakers from under-represented regions, directly contributing to the scientific community's diversity goals.
    Granted budget: 15K Euros (2023), 12K Euros (2024), 12K Euros (2025), 5K Euros (2026).

- **SPIRIT Open Call** 2024
  EU Horizon Europe - https://www.spirit-project.eu
  - Co-authored the winning research proposal, defining the scientific objectives, technical methodology, and work plan.
    Granted budget: 200K Euros.

LIST OF PUBLICATIONS

- **JOURNAL PUBLICATIONS**
  - **S. Rossi**, E. Hynes, B. Dunphy, I. Viola, X. Zhou, A. Verhofstadt, C. Reichherzer, K. Curtis, D. Mevlevioglu, A. Smolic, W. Van den Broeck, N. Murray, and P. Cesar. "*A Cookbook of User Metrics in Extended Reality*". [*Under Revision*] ACM Computing Surveys, 2026,
  - X. Zhou, I. Viola, **S. Rossi**, and P. Cesar. "*Comparison of Visual Saliency for Dynamic Point Clouds: Task-free vs. Task-dependent*" in IEEE Transactions on Visualization and Computer Graphics, 2025.
  - **S. Rossi**, I. Viola, L. Toni, and P. Cesar. "*A Clustering Approach to Unveil User Similarities in 6-DoF Extended Reality Applications.*" ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM), 2024. doi:10.1145/3701734.
  - P. Gomes, **S. Rossi**, and L. Toni. "*AGAR: Attention Graph-RNN for Adaptative Motion Prediction of Point Clouds of Deformable Objects.*" ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM), 2024. doi:10.1145/3662183.
  - S. Lee, I. Viola, **S. Rossi**, Z. Guo, I. Reimat, K. Ławicka, A. Striner, P. Cesar. "*Designing and Evaluating a VR Lobby for a Socially Enriching Remote Opera Watching Experience.*" in IEEE Transactions on Visualization and Computer Graphics, 2024. doi:10.1109/TVCG.2024.3372081.
  - **S. Rossi**, C. Ozcinar, A. Smolic, and L. Toni "*Do Users Behave Similarly in VR? Investigation of the User Influence on the System Design.*" ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM), 2020. doi:10.1145/3381846.
- **BOOK CHAPTER**
  - **S. Rossi**, A. Guedes, and L. Toni "*Streaming and user behavior in omnidirectional videos.*" in Immersive video technologies, Elsevier (2023). doi:10.1016/B978-0-32-391755-1.00009-2.
- **CONFERENCE PUBLICATIONS**
  - K. Wylężek, I. Viola, **S.Rossi**, J. Jansen, T. Röggla, P. Cesar."The Influence of Context on Learning in a Social VR Historical Fashion Exhibition". Proceedings of the 2026 ACM International Conference on Interactive Media Experiences (2026). Association for Computing Machinery, New York, NY, USA, 75–99. doi:10.1145/3788851.3805025
  - M. De Fré, J. van der Hooft, J. Jansen, **S. Rossi**, T. Röggla, T. Wauters, F. De Turck, I. Viola, and P. Cesar. "WebRTC-Based Volumetric Video Conferencing: SFU Architecture Evaluation and Benchmarking". In Proceedings of the 36th Workshop on Network and Operating System Support for Digital Audio and Video (NOSSDAV '26). Association for Computing Machinery, New York, NY, USA, 29–35. doi:10.1145/3798065.3798070
  - S. van Damme, J. Jansen, **S. Rossi**, and P. Cesar. "From Individual QoE to Shared Mental Models: A Novel Evaluation Paradigm for Collaborative XR" in Proceedings of the 17th International Conference on Quality of Multimedia Experience (QoMEX), 2025. doi:10.1109/QoMEX65720.2025.11219992
  - J. Jansen, T. Roggla, **S. Rossi**, I. Viola, and P. Cesar. "*Open-Sourcing VR2Gather: A Collaborative Social VR System for Adaptive Multi-Party Real Time Communication*" in Proceedings of the 32nd ACM International Conference on Multimedia (MM), 2024. doi:10.1145/3664647.3685515.
  - **S. Rossi**, I. Viola, L. Toni, and P. Cesar. "*Extending 3-DoF Metrics to Model User Behaviour Similarity in 6-DoF Immersive Applications*" in Proceedings of the 14th Conference on ACM Multimedia Systems (MMSys), 2023. doi:10.1145/3587819.3590976.

- P. Gomes, **S. Rossi** and L. Toni. "*Explaining Hierarchical Features in Dynamic Point Cloud Processing*" in IEEE Picture Coding Symposium (PCS), 2022. doi:10.1109/PCS56426.2022.10018000
- **S. Rossi**, I. Viola, L. Toni, and P. Cesar. "*A new Challenge: Behavioural analysis of 6-DoF user when consuming immersive media*" in Proceedings of IEEE International Conference on Image Processing (ICIP), doi:10.1109/ICIP42928.2021.9506525.
- P. Gomes, **S. Rossi** and L. Toni. "*Spatio-Temporal Graph-RNN for Point Cloud Prediction*" in Proceedings of IEEE International Conference on Image Processing (ICIP), 2021. doi:10.1109/ICIP42928.2021.9506084.
- **S. Rossi**, F. De Simone, P. Frossard and L. Toni. "*Spherical clustering of users navigating in 360° content*" in IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2019. doi:10.1109/ICASSP.2019.8683854.

- **WORKSHOP AND POSTERS**
  - J. Williamson, **S. Rossi**, R. Johnstone, I. Viola, J. Williamson, T. Roggla, D. A. Shamma, P. Cesar. "Social Density and its Impact on Behaviour in Virtual Environments". In Proceedings of the 3rd International Workshop on Interactive eXtended Reality (IXR '25), October 27–28, 2025, Dublin, Ireland. ACM, 2025. doi:10.1145/3746269.3760421
  - V. Pradhan, **S. Rossi**, and P. Cesar. "Exploring Entropy-Based Solutions for Trajectory Prediction in Virtual Reality", 2025. In the 17th International Workshop on IMmersive Mixed and Virtual Environment Systems (MMVE '25), March 31-April 4, 2025, Stellenbosch, South Africa. ACM, New York, NY, USA, 7 pages. doi:10.1145/3712677.3720460
  - T. Roggla, D. Shamma, J. Williamson, I. Viola, **S. Rossi**, P. Cesar. "*A Platform for Collecting User Behaviour Data during Social VR Experiments Using Mozilla Hubs*" in Proceedings of the 16th International Workshop on Immersive Mixed and Virtual Environment Systems (MMVE), 2024. doi:10.1145/3652212.3652225.
  - **S. Rossi**, L. Toni, and P. Cesar. "*Correlation between entropy and prediction error in VR Head Motion Trajectories*" in Proceedings of the 2nd ACM Multimedia (MM) workshop on Interactive eXtended Reality (IXR), 2023. doi:10.1145/3607546.3616805.
  - **S. Rossi**, I. Viola, and P. Cesar "*Behavioural Analysis of Navigation Trajectories in a 6-DoF VR System*" in NWO ICT.OPEN, Utrecht (The Netherlands), 2023.
  - **S. Rossi**, I. Viola, and P. Cesar. "*Behavioural Analysis in a 6-DoF VR System: Influence of Content, Quality and User Disposition*" in Proceedings of the 1st ACM Multimedia (MM) Workshop on Interactive EXtended Reality (IXR), 2022. doi:10.1145/3552483.3556454.
  - **S. Rossi**, I. Viola, J. Jansen, S. Subramanyam, L. Toni, and P. Cesar. "*Influence of Narrative Elements on User Behaviour in Photorealistic Social VR*" in Proceedings of the 13th ACM International Workshop on Immersive Mixed and Virtual Environment Systems (MMVE), 2021. doi:10.1145/3458307.3463371.
  - **S. Rossi** and L. Toni. "*Understanding user navigation in immersive experience: an information-theoretic analysis*" in Proceedings of the 12th ACM International Workshop on Immersive Mixed and Virtual Environment Systems (MMVE), 2020. doi:10.1145/3386293.3397115.
  - **S. Rossi** and L. Toni. "*Navigation-aware adaptive streaming strategies for omnidirectional video*". In IEEE International Workshop on Multimedia Signal Processing (MMSP), 2017. doi:10.1109/MMSP.2017.8122230
  - **S. Rossi** and L. Toni, "*Predicting Users Navigation in VR*" in PhD Network Symposium on Artificial Intelligence, London (UK), 2017.

- **TECHNICAL REPORTS**
  - **M. Cole, R. Baas, F. Yildirim, N. v. Breda, S. Rossi, H. Salim, P. Melis**, "Immersive Technologies Chapter," *SURF Tech Trend Report 2026*, 2025. https://www.surf.nl/tech-trends
- **MEDIA & PRESS**
  - "Crafting Algorithms to Optimise 3D Experience" · Academic Positions · 2023 https://academicpositions.com/story/crafting-algorithms-to-optimise-3d-experience
  - "How do we use space in VR?" · CWI · 2023 www.cwi.nl/en/stories/how-do-we-use-space-in-vr/
  - "Meet Silvia Rossi: PhD alumni researcher in VR" · UCL Engineering · 2023 https://www.ucl.ac.uk/engineering/meet-silvia-rossi
- **INVITED TALKS AND PANELS**
  - "*From Individual Immersion to Shared Experiences in Social XR*", Keynote Speaker at the ACM SIGMM Europe Chapter Symposium, co-located with ACM MM 2025, Dublin, Ireland.
  - "*Understanding User Behaviour to Unlock the Potential of XR*", Invited Presentation at the 2023 SIGMM award for outstanding PhD thesis in multimedia computing, communications and applications, ACM MM (November 2023), Ottawa, Canada.
  - "*Multimodal Large Foundation Models*", Invited Panellist at ACM MM (November 2023), Ottawa, Canada.
  - "*Cross Media Cafè: Spatial Computing*", Invited Speaker at Cross Media Cafè (October 2023), Hilversum, The Netherlands (Online Video).
  - "*Exploring User Interactivity to unlock the potential of Social XR*", Invited Presentation at the EEE department of UCL (July 2023), London, United Kingdom.
  - "*Social XR: Volumetric Video for Remote Communication and Collaboration*", Invited Presentation at the National XR days (July 2023), Delft, The Netherlands.
  - "*Understanding user interactivity for the next-generation immersive communication*", Invited Presentation at the NTNU (March 2023), Trondheim, Norway.
  - "*Adaptive Streaming for Immersive Communications*", Invited technical presentation at the 57th FITCE Congress Delivery and Consumption of Digital Media (`www.youtube.com/watch?v=Um1BsjDciZI`), MediaCityUK (Salford - UK), September 2018.